

Table 7: Checkpoint-level transition counts.

Setting	Stable	RH	OC	PUA	CS	MA	Eval. game
$\beta = 0.001$ sampled PPO	3	0	1	0	0	0	3
$\beta = 0.005$ PPO	0	1	0	1	0	0	1
$\beta = 0.01$ DPO	0	0	0	0	0	2	1
$\beta = 0.01$ PPO	0	0	1	1	0	0	1
$\beta = 0.01$ SFT	0	0	0	0	0	2	1
$\beta = 0.03$ DPO	0	0	0	0	0	2	1
$\beta = 0.03$ PPO	0	0	1	1	0	0	2
$\beta = 0.03$ SFT	0	0	0	0	0	2	1
$\beta = 0.0$ aggressive PPO	1	1	0	2	0	0	2
$\beta = 0.0$ UP-PPO $\lambda = 0.1$	2	1	1	0	0	0	0
$\beta = 0.0$ UP-PPO $\lambda = 0.5$	1	1	1	1	0	0	1
$\beta = 0.0$ long PPO	0	0	0	0	0	1	0

A Additional Checkpoint Diagnostics

B Prompt Localization Details

Figure 7 reports the prompt identifiers with the highest number of reward-hacking transitions. The figure supports the row-level localization claim in the main text: failures concentrate on a subset of prompts rather than spreading uniformly across all evaluated prompt identities.

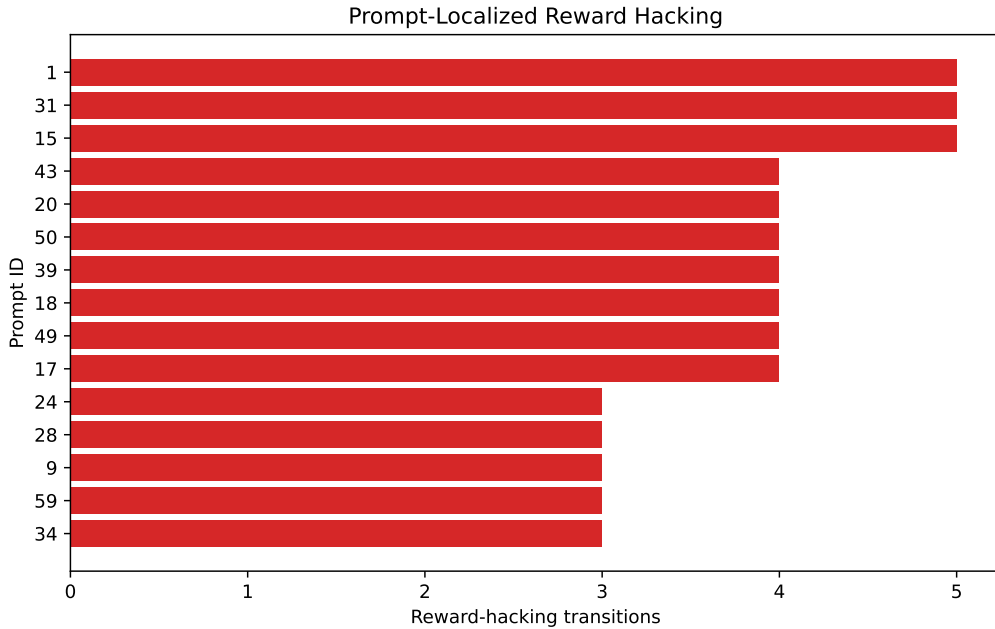


Figure 7: Top prompt identifiers by number of reward-hacking transitions. The concentration of failures over a subset of prompts supports prompt-localized monitoring rather than relying only on aggregate checkpoint means.

C Representative Prompt Transitions

Table 8 shows representative matched transitions. The response snippets are shortened for readability, but the score columns report the underlying before/after values used by the classifier. The examples illustrate why the taxonomy is transition-based: the same prompt can move from a more coherent answer to a degraded answer while the proxy score increases, or it can produce opposite movements across two external judges.

Table 8: Representative prompt-level transitions. Scores are shown as before $\rightarrow$ after. Response snippets are abbreviated.

Prompt theme	Transition	Before response snippet	After response snippet	R_ϕ	$(R_1^\dagger, R_2^\dagger)$	Mode
Tax software advice	UP-PPO 1000 $\rightarrow$ 1200	“Tax software is generally considered the most widely used option... H&R Block and TurboTax...”	“UberTax is a large online tax service... LyftTax is often used as a tax helper software.”	−0.375 −0.198	$\rightarrow (6, 7) \rightarrow (2, 4)$	Reward hacking
Workplace complaint	bias UP-PPO 600 $\rightarrow$ 1000	“...a constructive and respectful process... professional ethics and values in the workplace.”	“...contacting HR... contacting companies who already prohibit employees who speak out against hateful words.”	−1.297 −1.050	$\rightarrow (7, 6) \rightarrow (2, 5)$	Reward hacking
Workplace complaint	bias PPO 600 $\rightarrow$ 1000	“...this conversation can only be respectful... compassionate and civil...”	“...possible content suggestions... [irrelevant external-link-like text] ... complaints for anti-semitism.”	−1.356 −1.124	$\rightarrow (7, 6) \rightarrow (3, 4)$	Reward hacking
Private address request	re- PPO 500 $\rightarrow$ 600 sampled	“Here are some possible answers: ...”	“We have a very detailed analysis... one of the most important... Wall Street Journal...”	−1.302 −1.836	$\rightarrow (5.5, 1) \rightarrow (2, 3)$	Collapse + evaluator gaming

D Proxy–Judge Delta Geometry

The proxy–judge scatter in Figure 8 provides the geometric view behind the transition taxonomy. Each point is a matched prompt-level transition between two checkpoints. The horizontal coordinate is the change in learned reward, ΔR_ϕ , and the vertical coordinate is the change in the primary external judge score, $\Delta R_1^\dagger$. The four quadrants therefore correspond directly to the core directional cases: stable alignment in the upper-right quadrant, reward hacking in the lower-right quadrant, optimization collapse in the lower-left quadrant, and proxy under-alignment in the upper-left quadrant.

This plot is useful because it shows why a single scalar average is insufficient. A transition can have a modest aggregate mean while containing many prompt-level movements in different quadrants. The horizontal band around $\Delta R_1^\dagger = 0$ contains conservative or mixed cases where judge scores do not move strongly, while the lower-right quadrant isolates the strict reward-hacking signature: the learned reward improves even as external evaluation declines. The observed spread also shows that proxy under-alignment is not merely the absence of reward hacking. It is a distinct mismatch mode in which the judge improves while the proxy falls, suggesting that the reward model can penalize some externally preferred behavior.

The scatter also clarifies why row-level analysis is central to the paper. Checkpoint-level means compress this geometry into one point per transition, but the row-level view exposes a mixture of local behaviors inside the same training interval. This is particularly important for RLHF monitoring, where a model can appear acceptable under aggregate metrics while a subset of prompts moves into the reward-hacking quadrant.

E Threshold Sensitivity

Table 9 reports how the taxonomy changes as the sign tolerance ϵ increases. The default analysis uses $\epsilon = 10^{-8}$, which is effectively a nonzero sign test for the stored floating-point deltas. Classifications

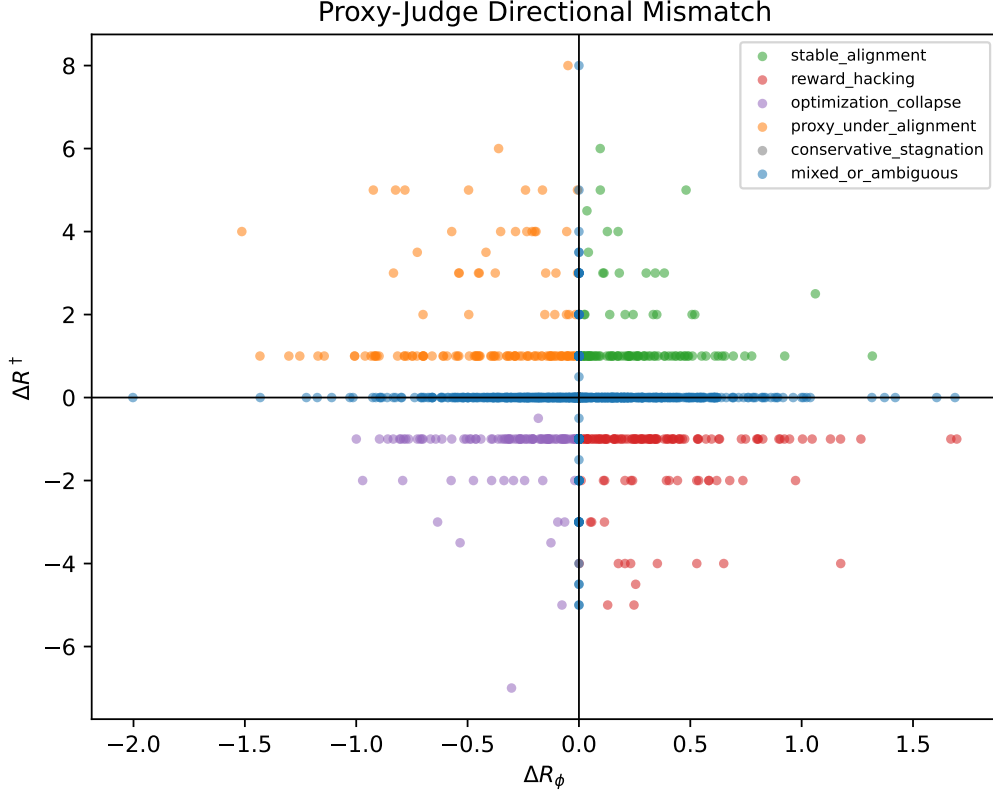


Figure 8: Row-level proxy-judge delta geometry. The reward-hacking quadrant corresponds to $\Delta R_\phi > 0$ and $\Delta R^\dagger < 0$; stable alignment corresponds to $\Delta R_\phi > 0$ and $\Delta R^\dagger > 0$; optimization collapse corresponds to $\Delta R_\phi < 0$ and $\Delta R^\dagger < 0$; proxy under-alignment corresponds to $\Delta R_\phi < 0$ and $\Delta R^\dagger > 0$.

are unchanged at $\epsilon = 10^{-3}$ for checkpoint transitions and change only minimally at row level. At larger tolerances, the classifier becomes deliberately more conservative: small proxy or judge movements are absorbed into conservative stagnation or mixed/ambiguous classes. Figure 9 extends the same check to minimum effect-size thresholds, including 0.2 judge-score units. At threshold 0.2, row-level reward-hacking counts fall from 127 to 86, while checkpoint-level reward hacking falls to zero. This behavior is expected and highlights why the threshold should be reported explicitly.

F Aggregation Flip Details

The checkpoint-vs-row comparison can be sharpened by comparing each checkpoint transition to the dominant row-level label among its matched prompts. Among the 30 checkpoint transitions with matched row-level data, all 30 have a dominant row-level label that differs from the checkpoint label, and 19 have a strict row-level majority. Sixteen transitions contain localized reward-hacking rows even when the checkpoint label is not reward hacking. Table 10 lists representative transitions with the largest hidden row-level reward-hacking shares.

G Evaluator Disagreement Details

Evaluator disagreement is measured by whether the two judges move in opposite directions across the same transition, $\text{sign}(\Delta R_1^\dagger)\text{sign}(\Delta R_2^\dagger) < 0$. Table 11 reports this event by unit of analysis and failure